Ben Phung

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EDUCATION

Harvey Mudd College **May 2028**

Bachelor of Science, Engineering

• **Achievements:** Dean's List honors

• **Coursework:** Design and Manufacturing, Systems, Solid and Fluid Mechanics, Computer Engineering/Digital Design, Experimental Engineering, Computer Science, Environmental Economics, Micro-Processors

PROFESSIONAL EXPERIENCE

Data Pillar - *Engineering Intern* **May 2025 - Aug 2025**

• Developed Python-based tools to automate data processing tasks, reducing manual effort in internal workflows.

• Collaborated with engineers to develop the data pipeline using C++ modules and improve workflow efficiency.

• Analyzed datasets using MATLAB to inform backend feature development and product improvements.

• Contributed to client-driven product improvements by synthesizing feedback from meetings and proposing actionable enhancements.

PROJECTS

Sponsored Project with Partnerships for Global Surgery - *Electromechanical Engineering* **Aug 2026 - Present**

• Collaborate with a multidisciplinary clinical and engineering team to develop a reusable pediatric bronchoscopy device for low-resource hospitals, with a system cost under $10,000 and a service life of 100+ uses

• Design the mechanical housing and camera integration for a miniature endoscope, packaging a 2.2 mm camera inside a 2.5 mm outer-diameter titanium housing while leaving room for surgical instruments and ventilation

• Evaluate optical and mechanical design trade-offs for airway visualization, including a 120-degree field of view, viewing-direction adjustment, LED illumination, protective window sealing, and lens-cleaning methods

Underwater Robot - Experimental Engineering **Jan 2026 - Present**

- Designed analog signal conditioning circuits using KiCad schematic capture for op-amp amplifiers and filters, achieving 0.3–3.3V sensor scaling for Teensy firmware integration.
- Designed and 3D printed custom dividers to house the motherboard, battery, wiring, and other components to prevent short circuits and improve interchangeability.
- Conducted sensor calibration and hardware bring-up, using oscilloscopes and data acquisition with MATLAB to debug and validate system performance.
- Performed CFD simulations in COMSOL and validated results against wind tunnel data to inform hull material selection, balancing hydrodynamic performance with budget constraints.

Filament Waste Collection System - Engineering Design and Manufacturing Project **Jan 2025 - May 2025**

- Partnered with the HMC Makerspace to design and fabricate a centralized filament waste collection system for the 3D printing lab, which reduced time to dispose of material waste by ~50%.
- Led a 5-person team through iterative prototyping using SolidWorks, 3D printing, and machining equipment (lathe, mill) to develop an effective and affordable manufacturing process for our client.
- Managed project budget (BOM), schedule, and client deliverables, progressing from concept through final manufacturing and deployment.

ACTIVITIES / SERVICES

Mudd Sub - Autonomous Underwater Vehicle - *PCB Team Lead* **Jan 2026 - Present**

- Designed and optimized custom PCBs for an AUV, improving reliability and modularity through iterative layout in KiCad and Altium and validating with bench-level hardware testing.
- Designed and manufactured the structural chassis of an autonomous underwater vehicle (AUV), optimizing for watertight sealing, buoyancy, and electrical component integration using CAD and machining processes.

Adaptive Design at Mudd (ADAM) - *Design Engineer* **Jan 2024 - Present**

• Design and fabricate custom assistive devices for individuals with disabilities and elderly community members by iteratively prototyping using CAD and 3D printing.

• Built assistive tools, including a GIC checkerboard and 3D-printed wheelchair components.

TECHNICAL SKILLS

• **Design and Manufacturing**: CAD (SolidWorks), 3D Printing, Laser Cutting, Welding, Metal/Woodworking, CNC

• **Programming**: Python, C/C++, SystemVerilog

• **Hardware/Microcontrollers**: Arduino, Raspberry Pi. STM32 (ARM Cortex-M)

• **Interfaces & Architecture:** RISC-V, SPI, UART

• **Software**: Microsoft Office Suite

• **Simulation and Analysis**: MATLAB, COMSOL, KiCAD, Altium, LTspice